

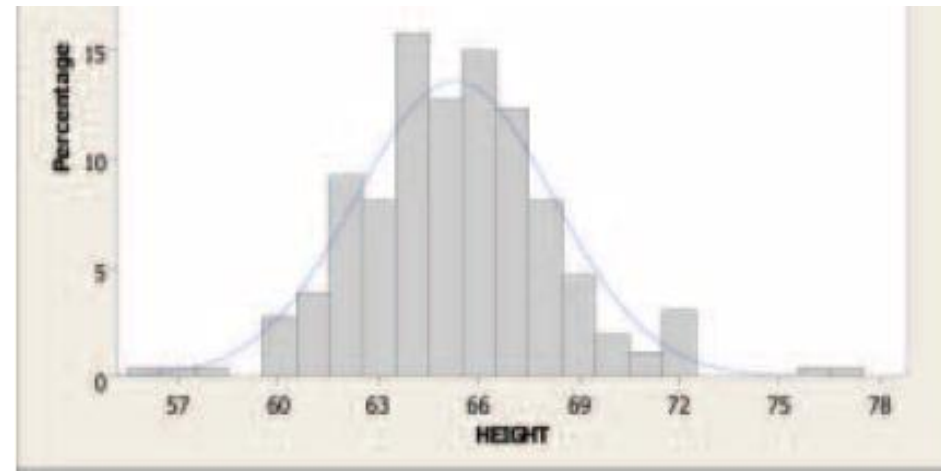
Inferential Statistics

Continuous Distributions

CSDA 5110

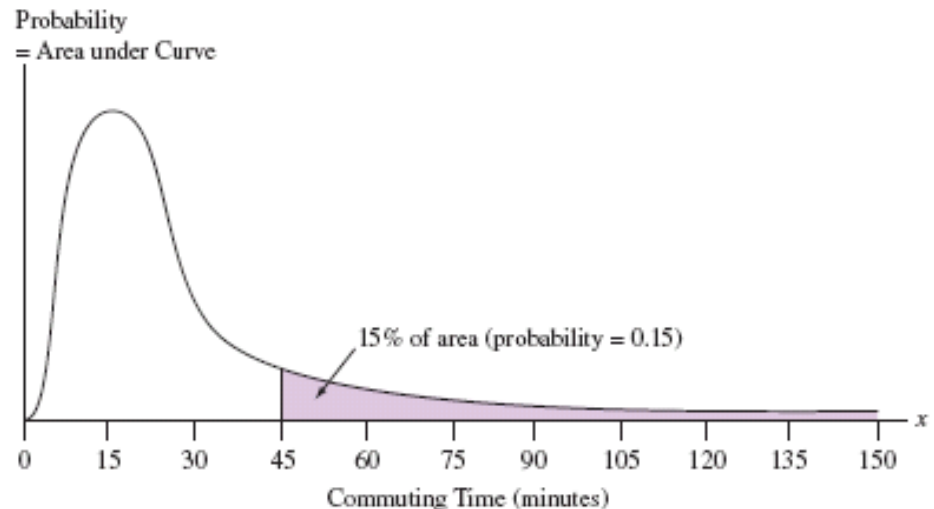
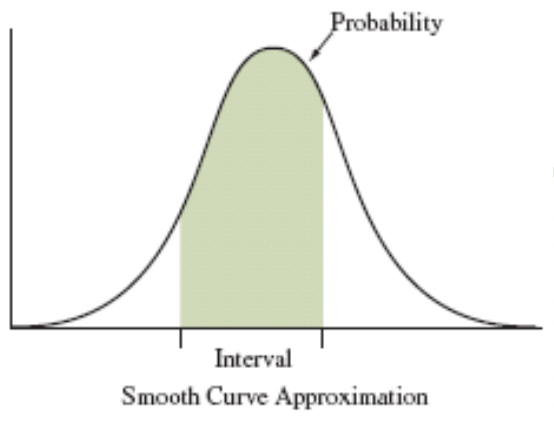
Continuous Random Variable

- A continuous random variable has an infinite continuum of possible values in an interval.
- Examples are: time, age and size measures such as height and weight.
- Continuous variables are measured in a discrete manner because of rounding.



Probability Distribution of a Continuous Random Variable

- A continuous random variable has possible values that form an interval.
- Its probability distribution is specified by a curve.
- Each interval has probability between 0 and 1.
- The interval containing all possible values has probability equal to 1.



Probability Distribution

For every quantitative data:

Always plot your data: make a graph

- Look for overall pattern (shape, center, spread. outliers)
- Choose either five number summary or mean and standard deviation to describe center and spread.
- Sometimes the overall pattern of a large number of observations is so regular that we can describe it by smooth curve.

Normal Distribution

The normal distribution is symmetric, bell-shaped and characterized by its mean μ and standard deviation σ .

- The normal distribution is the most important distribution in statistics
 - Many distributions have an approximate normal distribution
 - Approximates many discrete distributions well when there are a large number of possible outcomes
 - Many statistical methods use it even when the data are not bell shaped

Normal Distribution

Normal distributions are

- Bell shaped
- Symmetric around the mean

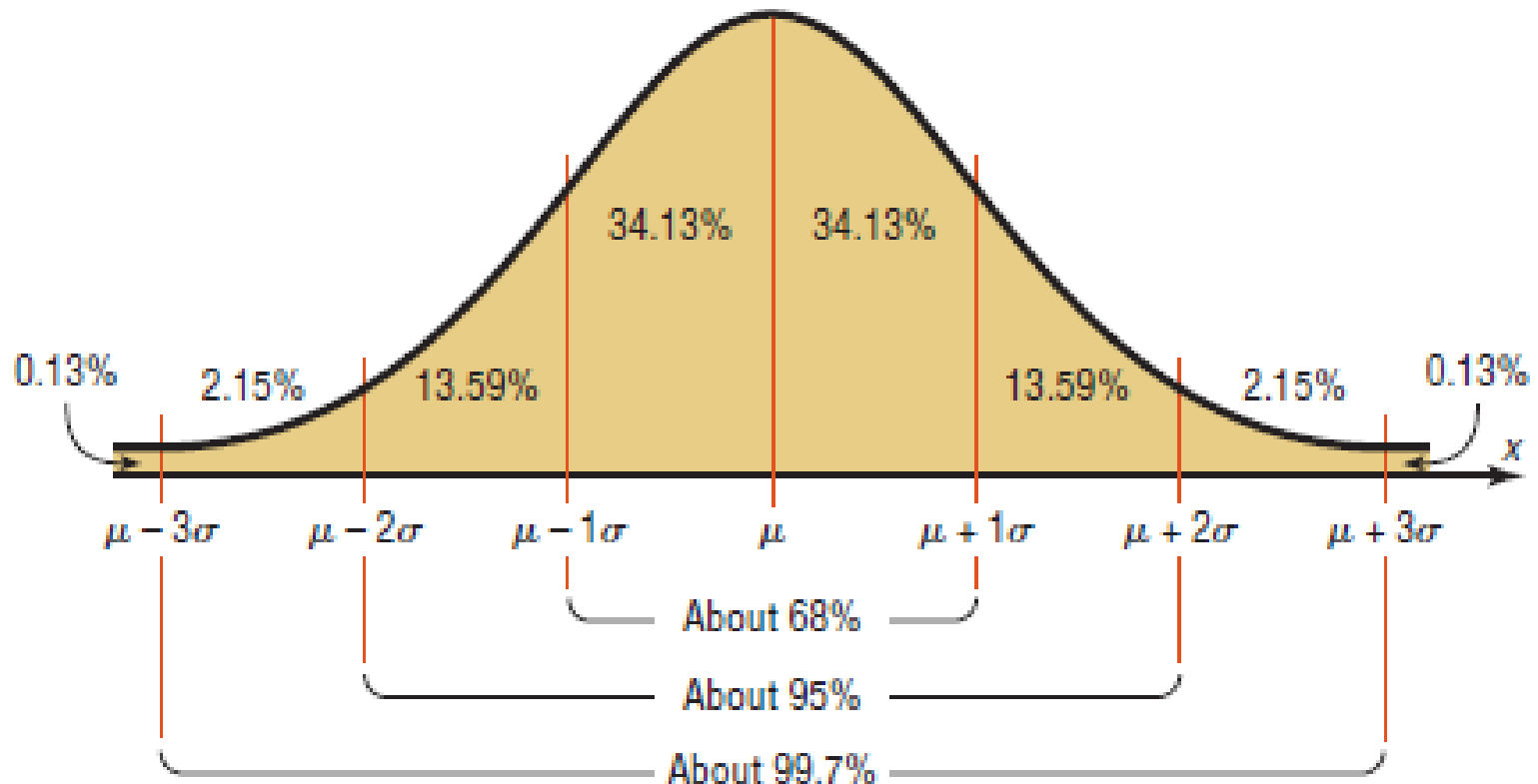
The mean (μ) and the standard deviation (σ) completely describe the density curve

- Increasing/decreasing μ moves the curve along the horizontal axis
- Increasing/decreasing σ controls the spread of the curve

Empirical Rule for Any Normal Curve

- **68%** of the observations fall within one standard deviation of the mean
- **95%** of the observations fall within two standard deviations of the mean
- **99.7%** of the observations fall within three standard deviations of the mean

Normal Distribution Properties



EXAMPLE : Cell Phone Rates

Monthly charges for cell phone plans in the United States are normally distributed with mean $\mu = \$62$ and standard deviation $\sigma = \$18$.

- What proportion of plans charge less than \$44?
- What proportion of plans charge more than \$98?
- What proportion of plans charge between \$44 and \$98?
- What are the charges between which 95% of the plans fall?

Z-Scores and the Standard Normal Distribution

- A standard normal distribution has mean $\mu=0$ and standard deviation $\sigma=1$
- When a random variable has a normal distribution and its values are converted to z-scores by subtracting the mean and dividing by the standard deviation, the z-scores have the standard normal distribution.

$$P(X < x)$$

- Adult systolic blood pressure is normally distributed with $\mu = 120$ and $\sigma = 20$. What percentage of adults have systolic blood pressure less than 100?
- $P(X < 100) = \text{pnorm}(100, 120, 20)$ (Rcode)
= .1587

$$P\left(Z < \frac{(100 - 120)}{20}\right) = P(Z < -1.00) = .1587$$

$$\text{pnorm}(-1, 0, 1) \text{ (Rcode)}$$

$$P(X > x)$$

- Adult systolic blood pressure is normally distributed with $\mu = 120$ and $\sigma = 20$. What percentage of adults have systolic blood pressure greater than 133?

- $P(X > 133) = 1 - P(X < 133)$
 $= 1 - \text{pnorm}(133, 120, 20)$ (Rcode)
 $= .2578$

$$P\left(Z < \frac{(133 - 120)}{20}\right) = P(Z < .65) = .7422$$

- $1 - \text{pnorm}(.65, 0, 1)$
- 25.8% of adults have systolic blood pressure greater than 133

$P(a < X < b)$ is same as Area between two values

Adult systolic blood pressure is normally distributed with $\mu = 120$ and $\sigma = 20$. What percentage of adults have systolic blood pressure between 100 and 133?

- $$\begin{aligned} P(100 < X < 133) &= P(X < 133) - P(X < 100) \\ &= (\text{pnorm}(133, 120, 20) - \text{pnorm}(100, 120, 20)) \\ &= .5835 \end{aligned}$$

$$P\left(Z < \frac{(133 - 120)}{20}\right) - P\left(Z < \frac{(100 - 120)}{20}\right) =$$

$$P(Z < .65) - P(Z < -1.00) = .7422 - .1587 = .5835$$

$$(\text{pnorm}(.65, 0, 1) - \text{pnorm}(-1, 0, 1)) .5835$$

58% of adults have systolic blood pressure between 100 and 133

Find X Value Given Area to Left

Adult systolic blood pressure is normally distributed with $\mu = 120$ and $\sigma = 20$. What is the 1st quartile?

- $P(X < x) = .25$, find x :
 - `qnorm(0.25, mean = 120, sd = 20) = 106.51`

Find X Value Given Area to Right

Adult systolic blood pressure is normally distributed with $\mu = 120$ and $\sigma = 20$.

10% of adults have systolic blood pressure above what level?

- $P(X > \underline{x}) = .10$, means $P(X < x) = .90$
 $qnorm(0.90, \text{mean} = 120, \text{sd} = 20) = 145.63$

t-Distribution

- In practice, we don't know the population standard deviation σ
- Substituting the sample standard deviation s for σ to get $se = s/\sqrt{n}$ introduces extra error
- To account for this increased error, we replace the *z-score* by a slightly larger score, the *t-score*
- These *t*- values follow the **Student *t* distribution**, most often called the ***t* distribution**.

Characteristics of the t Distribution

The t distribution is similar to the standard normal distribution in these ways:

1. It is bell-shaped.
2. It is symmetric about the mean.
3. The mean, median, and mode are equal to 0 and are located at the center of the distribution.
4. The curve never touches the x axis.

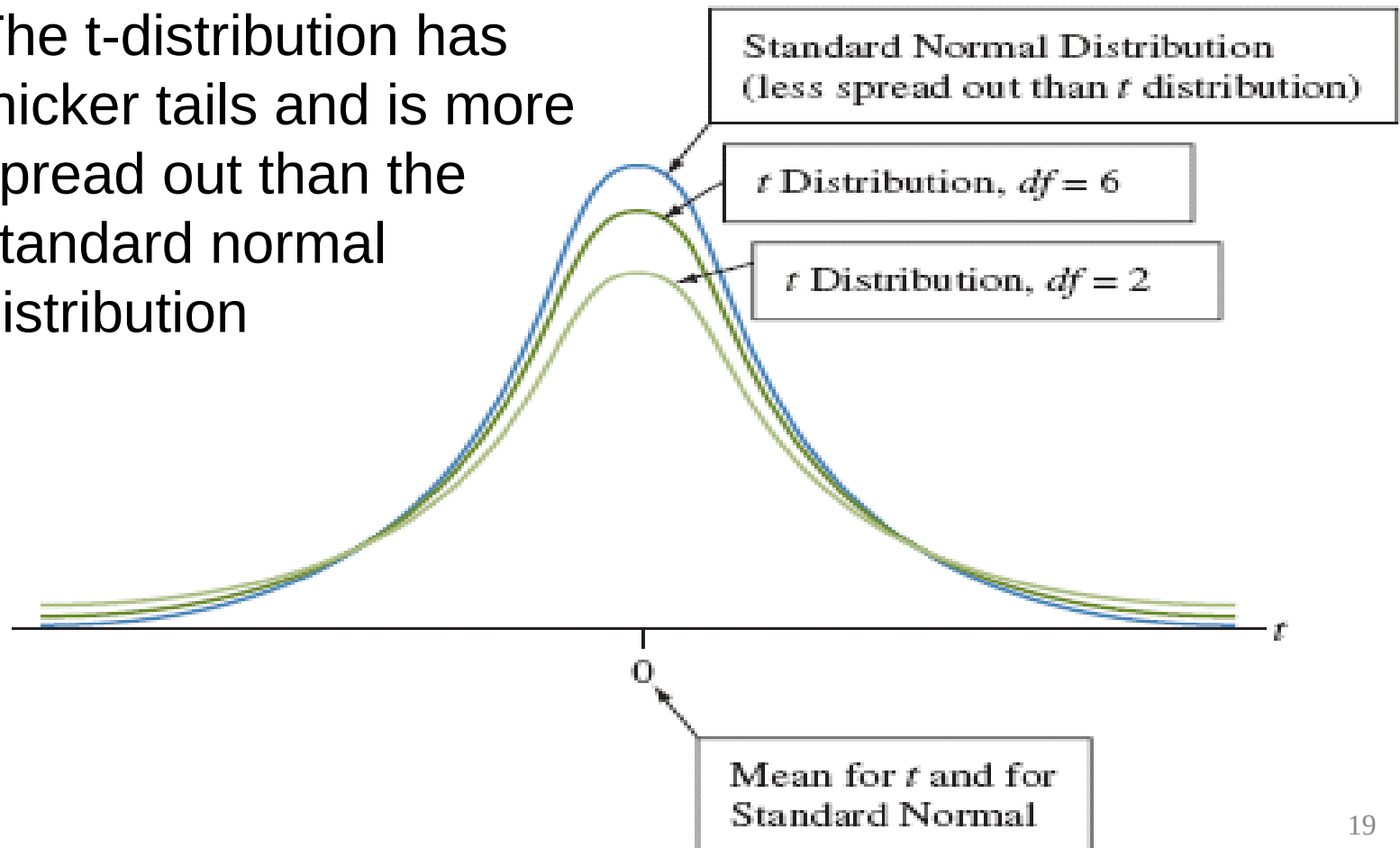
Characteristics of the t Distribution

The t distribution differs from the standard normal distribution in the following ways:

1. The variance is greater than 1.
2. The t distribution is actually a family of curves based on the concept of **degrees of freedom**, which is related to sample size.
3. As the sample size increases, the t distribution approaches the standard normal distribution.

t Distribution

The t-distribution has thicker tails and is more spread out than the standard normal distribution



Finding area for t-distribution

- For sample size $n=15$,
find $\text{Prob}(X < 2) = \text{pt}(2, 14) = .967$ (Rcode)
 $\text{df}=14$
- For sample size $n=15$,
- Find x when area to the right is $.05$
 $\text{qt}(0.95, 14) = 1.761$